SINGLE-GPU LORA RANK ABLATION

Brent Oatis | Applied ML Engineering | Short-horizon cooperative demand forecasting

DECISION: Retain rank 4 for this tested configuration. Five-seed error ranges overlap, so the evidence supports an efficiency choice - not an accuracy-superiority claim.

MEASURED RESULTS

LoRA rank	Runs	Test MAPE mean +/- SD	Seed range	Trainable params	Peak GPU	Adapt + eval mean +/- SD
4	5	3.2396% +/- 0.0381	3.1860-3.2931%	2,048	123.02 MB	14.19 +/- 0.14 s
8	5	3.2487% +/- 0.0466	3.1854-3.3143%	4,096	123.42 MB	13.94 +/- 0.23 s
16	5	3.2669% +/- 0.0578	3.1825-3.3333%	8,192	124.21 MB	13.94 +/- 0.13 s

Hardware footnote: NVIDIA Tesla T4, 15,360 MiB VRAM; driver 580.82.07; host CUDA 13.0; PyTorch 2.9.1+cu128 using CUDA 12.8; FP32; batch 128; 48-hour input; six-hour horizon; shared-backbone pretraining 53.82 s.

QUESTION AND METHOD

Does rank 4, 8, or 16 change forecast error when data, chronological split, architecture, shared pretrained backbone, optimizer, epochs, precision, batch size, and evaluation stay fixed? Fifteen confirmatory adaptations used seeds 101, 202, 303, 404, and 505 for each rank.

EVIDENCE AND REPRODUCIBILITY

Pinned direct dependencies, deterministic settings, a YAML-driven single entry point, restartable per-run JSON records, raw and summary CSVs, a plot generated from raw data, dataset SHA-256 provenance, full environment capture, and automated integrity checks. The Week 8 package reuses the 15 preserved Week 7 measurements; it does not invent or impute results.

INTENDED USE

Portfolio evidence of controlled experimentation, memory/compute budgeting, honest benchmarking, and auditable model documentation. It may guide the design of a real-data pilot after approval and validation.

BOUNDED INTERPRETATION

Rank 4 had the lowest mean MAPE, but the difference from rank 8 was only 0.0092 percentage points and every rank's seed range overlapped. This is descriptive evidence from one small model, one generated dataset, one shared pretraining seed, and one GPU class - not proof of equivalence or a universal rank optimum.

KEY LIMITATIONS

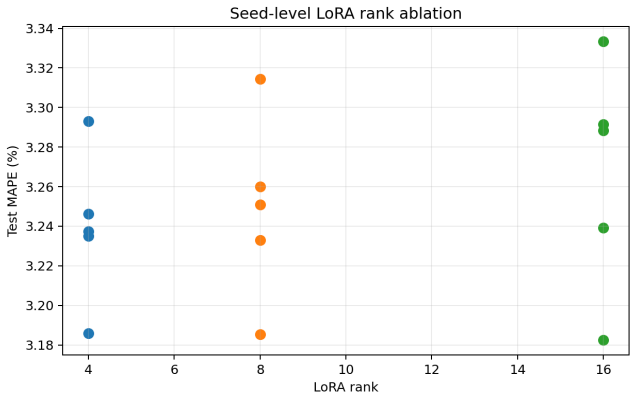
Synthetic demand is not representative customer data; pretraining and data-generation seeds were not repeated; ranks were not tuned separately; no real-cooperative, extreme-weather, subgroup, drift, or out-of-distribution validation was performed. Do not use this model for dispatch, planning, billing, or safety-critical decisions.

GOVERNANCE POSTURE

Intended use, provenance, configuration, evaluation records, human decision gates, and residual risk map to NIST AI RMF, ISO/IEC 42001, and EU Annex IV-style documentation. The package claims no certification or production readiness.

INDUSTRY TARGET

Applied Machine Learning Engineer supporting forecasting and grid analytics at Great River Energy. The artifact demonstrates how to turn a one-GPU experiment into a reviewable decision record for reliability-focused work serving member-owner cooperatives.



PORTFOLIO EVIDENCE

Controlled design
Rank-only ablation with a shared backbone and preregistered decision rule.

Honest measurement
Five seeds per rank; raw values, units, mean, sample SD, range, time, and memory.

Reproducibility engineering
Pinned environment, automatic capture, deterministic settings, restartable CLI, and integrity tests.

Model governance
Provenance, intended use, exclusions, limitations, residual risk, and Annex IV-style records.

Next evidence
Real approved data, repeated pretraining/data seeds, rank-specific tuning, stress slices, drift, and target-system validation.

PORTFOLIO LINKS

Repository: github.com/brentoatis-collab/cst627-lora-rank-ablation-portfolio

Recorded walkthrough: LINK PENDING AFTER RECORDING